

P6: Using Learning to Create a Novel Game Controller

Section 1: Setup and Train the Initial Network

Objective: Achieve 60%+ accuracy on <150k params

Executed by: Shah, Saurav

Initial Network

Model: "sequential"		
Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 148, 148, 16)	448
max_pooling2d (MaxPooling2D)	(None, 74, 74, 16)	0
conv2d_1 (Conv2D)	(None, 72, 72, 32)	4,640
max_pooling2d_1 (MaxPooling2D)	(None, 36, 36, 32)	0
conv2d_2 (Conv2D)	(None, 34, 34, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 17, 17, 64)	0
conv2d_3 (Conv2D)	(None, 15, 15, 32)	18,464
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 32)	0
flatten (Flatten)	(None, 1568)	0
dense (Dense)	(None, 64)	100,416
dropout (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 3)	195
Total params: 142,659 (557.26 KB)		
Trainable params: 142,659 (557.26 KB)		
Non-trainable params: 0 (0.00 B)		

Total Params: 142,659 (<150k)

Loss Plot

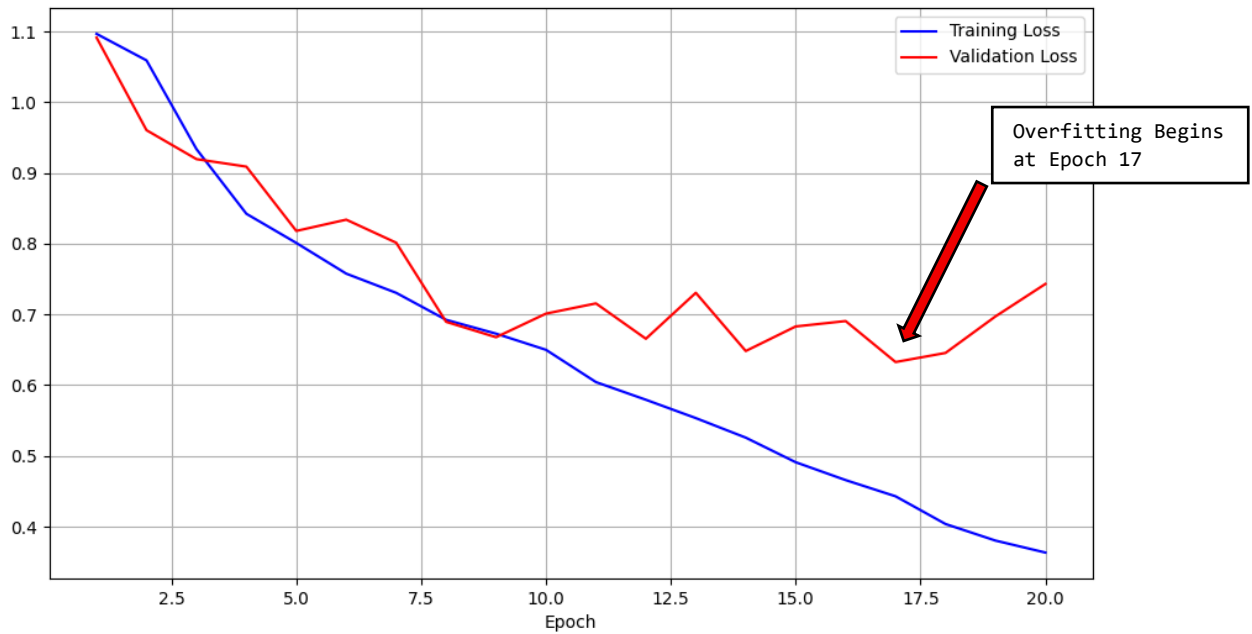


Fig 1. Training and Validation Loss

Result:

```
Epoch 16/20
32/32 ————— 6s 171ms/step - accuracy: 0.8215 - loss: 0.4658 - val_accuracy: 0.7150
- val_loss: 0.6906
Epoch 17/20
32/32 ————— 6s 172ms/step - accuracy: 0.8290 - loss: 0.4429 - val_accuracy: 0.7330
- val_loss: 0.6327
Epoch 18/20
32/32 ————— 6s 171ms/step - accuracy: 0.8512 - loss: 0.4037 - val_accuracy: 0.7530
- val_loss: 0.6456
Epoch 19/20
32/32 ————— 6s 172ms/step - accuracy: 0.8560 - loss: 0.3803 - val_accuracy: 0.7300
- val_loss: 0.6972
```

Accuracy Plot

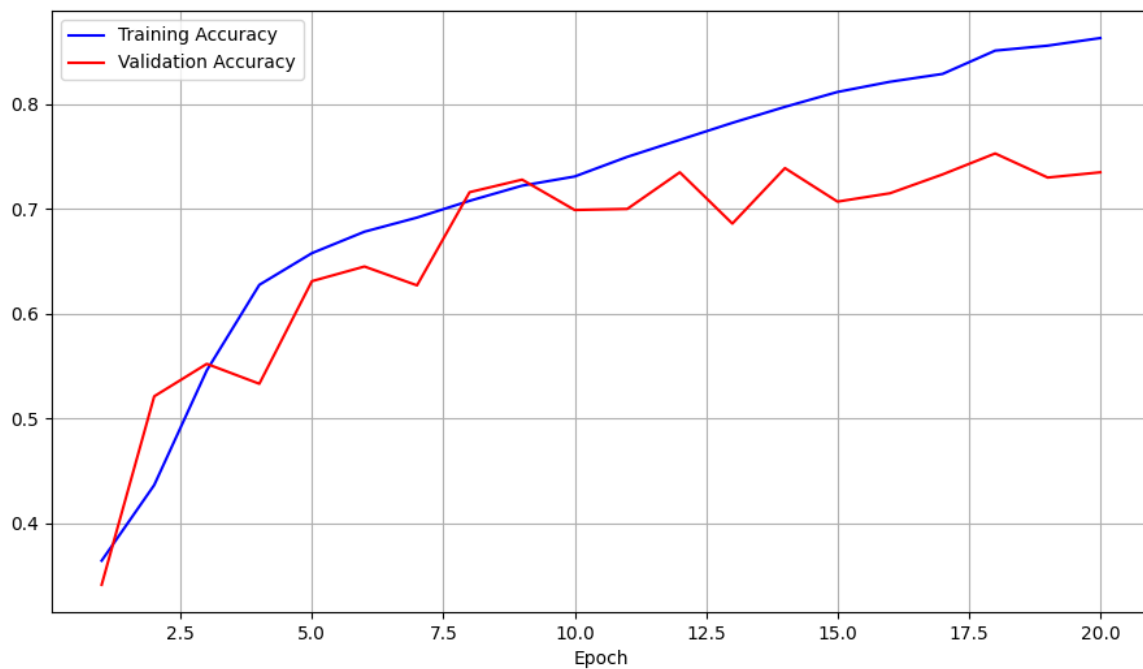


Fig 2. Training and Validation Accuracy

Test Set Evaluation

```
* Evaluating basic_model  
30/30 ————— 2s 52ms/step - accuracy: 0.7358 - loss: 0.7385
```

Test Accuracy: 73.58% (required 60%+)

Best validation accuracy: 75.3% at epoch 18

Overfitting point: epoch 17 (lowest validation loss: 0.6327)